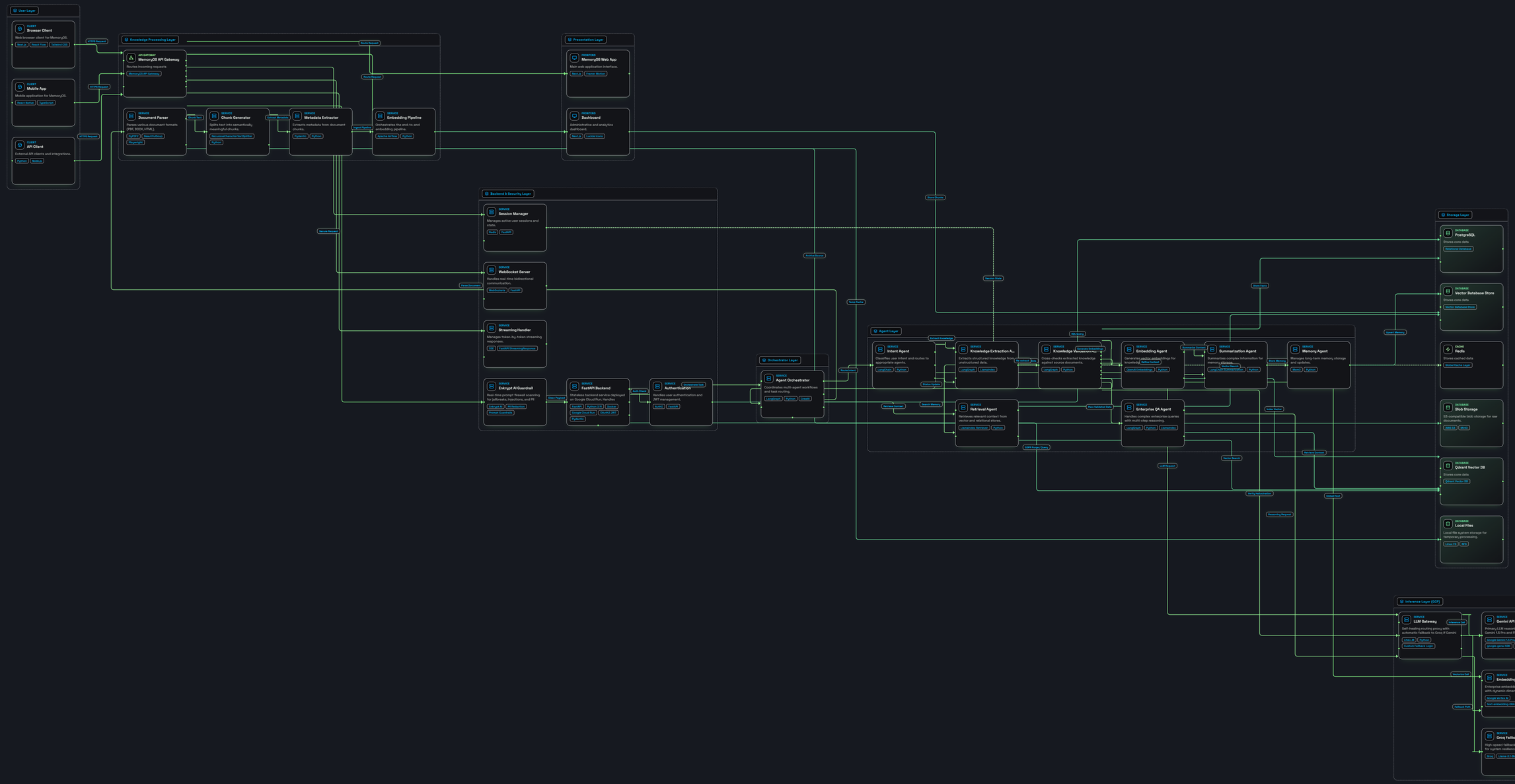

User Layer
CLIENT
Browser Client
Web browser client for MemoryOS.
Next.js
React Flow
Tailwind CSS
CLIENT
Mobile App
Mobile application for MemoryOS.
React Native
TypeScript
CLIENT
API Client
External API clients and integrations.
Python
Node.js
HTTPS Request
HTTPS Request
HTTPS Request
Knowledge Processing Layer
API GATEWAY
MemoryOS API Gateway
Routes incoming requests
MemoryOS API Gateway
Route Request
Route Request
SERVICE
Document Parser
Parses various document formats (PDF, DOCX, HTML).
PyPDF2
BeautifulSoup
Playwright
Chunk Text
SERVICE
Chunk Generator
Splits text into semantically meaningful chunks.
RecursiveCharacterTextSplitter
Python
Extract Metadata
SERVICE
Metadata Extractor
Extracts metadata from document chunks.
Pydantic
Python
Ingest Pipeline
SERVICE
Embedding Pipeline
Orchestrates the end-to-end embedding pipeline.
Apache Airflow
Python
Presentation Layer
FRONTEND
MemoryOS Web App
Main web application interface.
Next.js
Framer Motion
FRONTEND
Dashboard
Administrative and analytics dashboard.
Next.js
Lucide Icons
Secure Request
Parse Document
Backend & Security Layer
SERVICE
Session Manager
Manages active user sessions and state.
Redis
FastAPI
SERVICE
WebSocket Server
Handles real-time bidirectional communication.
WebSockets
FastAPI
SERVICE
Streaming Handler
Manages token-by-token streaming responses.
SSE
FastAPI StreamingResponse
SERVICE
Enkrypt AI Guardrail
Real-time prompt firewall scanning for jailbreaks, injections, and PII
Enkrypt AI
PII Redaction
Prompt Guardrails
Clean Payload
SERVICE
FastAPI Backend
Stateless backend service deployed on Google Cloud Run. Handles
FastAPI
Python 3.11
Docker
Google Cloud Run
OAuth2 JWT
Pydantic
Auth Check
SERVICE
Authentication
Orchestrate Task
Handles user authentication and JWT management.
Auth0
FastAPI
Store Chunks
Archive Source
Session State
Temp Cache
Orchestrator Layer
SERVICE
Agent Orchestrator
Coordinates multi-agent workflows and task routing.
LangGraph
Python
CrewAI
Route Intent
Agent Layer
Extract Knowledge
SQL Query
SERVICE
Intent Agent
Classifies user intent and routes to appropriate agents.
LangChain
Python
Status Update
SERVICE
Knowledge Extraction A...
Extracts structured knowledge from unstructured data.
LangGraph
LlamaIndex
Re-extract Data
SERVICE
Knowledge Validation A...
Generate Embeddings
Cross-checks extracted knowledge against source documents.
LangGraph
Python
SERVICE
Embedding Agent
Generates vector embeddings for knowledg
Refine Context
OpenAI Embeddings
Python
Summarize Context
SERVICE
Summarization Agent
Summarizes complex information for memory storage.
Vector Search
LangChain Summarization
Python
Store Memory
SERVICE
Memory Agent
Manages long-term memory storage and updates.
Mem0
Python
Store Facts
Upsert Memory
Retrieve Context
Search Memory
SERVICE
Retrieval Agent
Retrieves relevant context from vector and relational stores.
LlamaIndex Retriever
Python
Pass Validated Data
SERVICE
Enterprise QA Agent
Handles complex enterprise queries with multi-step reasoning.
LangGraph
Python
LlamaIndex
Index Vector
GDPR Purge / Query
LLM Request
Vector Search
Retrieve Context
Verify Hallucination
Embed Text
Reasoning Request
Storage Layer
DATABASE
PostgreSQL
Stores core data
Relational Database
DATABASE
Vector Database Store
Stores core data
Vector Database Store
CACHE
Redis
Stores cached data
Global Cache Layer
DATABASE
Blob Storage
S3-compatible blob storage for raw documents.
AWS S3
MinIO
DATABASE
Qdrant Vector DB
Stores core data
Qdrant Vector DB
DATABASE
Local Files
Local file system storage for temporary processing.
Linux FS
NFS
Inference Layer (GCP)
SERVICE
LLM Gateway
Inference Call
Self-healing routing proxy with automatic fallback to Groq if Gemini
LiteLLM
Python
Custom Fallback Logic
SERVICE
Gemini API
Primary LLM reasoning layer using Gemini 1.5 Pro and Flash.
Google Gemini 1.5 Pro/Flash
google-genai SDK
Google Cloud
Vectorize Call
SERVICE
Embedding API
Enterprise embedding generation with dynamic dimension handling
Google Vertex AI
text-embedding-004
Fallback Path
SERVICE
Groq Fallback API
High-speed fallback LLM endpoint for system resilience.
Groq
Llama-3.1-8b-instant